30Days SQL Micro Course Certificate Assignment

Dataset:-Hospital.csv

	hospital_name character varying (100)	location character varying (100)	department character varying (100)	doctors_count integer	patients_count integer	admission_date date	discharge_date date	medical_expenses numeric (10,2)
1	Wellness Clinic	Chennai	Orthopedics	47	182	2023-12-13	2023-12-14	31364.88
2	Fortis Care	Pune	ENT	15	51	2023-12-29	2024-01-09	47280.19
3	Wellness Clinic	Ahmedabad	Pediatrics	20	120	2023-10-07	2023-10-21	28574.72
4	Heritage Hospital	Hyderabad	Urology	8	172	2023-04-29	2023-05-11	7000.83
5	City Hospital	Kolkata	Gynecology	35	76	2023-02-10	2023-02-12	47210.46
6	Heritage Hospital	Hyderabad	Oncology	11	76	2023-02-05	2023-02-17	18612.34
7	Global Medicare	Ahmedabad	Oncology	22	99	2023-08-01	2023-08-06	47808.55
8	Apollo Health	Jaipur	General Medicine	37	173	2023-10-01	2023-10-12	12284.65
9	Heritage Hospital	Jaipur	ENT	9	198	2023-06-12	2023-06-18	14650.23
10	Green Valley Hospital	Pune	Pediatrics	30	107	2023-12-07	2023-12-08	30006.96
11	Healing Touch	Mumbai	Dermatology	39	70	2023-09-13	2023-09-15	13282.51
12	Fortis Care	Jaipur	Urology	22	38	2023-03-12	2023-03-25	13481.90
13	City Hospital	Lucknow	Cardiology	32	67	2023-11-18	2023-11-24	36748.42
14	Fortis Care	Jaipur	Orthopedics	27	60	2023-09-18	2023-09-24	37056.51
15	Sunrise Medical	Ahmedabad	Gynecology	48	37	2023-08-06	2023-08-11	15107.89
16	Healing Touch	Chennai	Cardiology	17	60	2023-06-27	2023-06-30	9348.55
17	Green Valley Hospital	Chennai	Pediatrics	40	96	2023-03-22	2023-03-28	48548.44
18	Healing Touch	Ahmedabad	Neurology	14	68	2023-11-11	2023-11-22	29942.75
19	Apollo Health	New Delhi	Urology	41	111	2023-11-05	2023-11-08	43367.53
20	Fortis Care	Mumbai	Urology	46	154	2023-02-14	2023-02-18	41367.10
21	Metro Hospital	Bengaluru	Dermatology	7	68	2023-10-06	2023-10-07	30136.26
22	Apollo Health	Hyderabad	Dermatology	28	191	2023-04-08	2023-04-22	17698.49
23	Fortis Care	Ahmedabad	Oncology	32	116	2023-10-28	2023-10-30	40799.02
24	Apollo Health	Hyderabad	Orthopedics	5	79	2023-03-25	2023-03-29	20952.51
25	City Hospital	Lucknow	Oncology	48	91	2023-02-24	2023-03-04	10882.40
26	Apollo Health	Pune	ENT	9	59	2023-11-09	2023-11-16	25177.79
27	Apollo Health	Lucknow	ENT	9	171	2023-04-06	2023-04-21	21666.78
28	Apollo Health	Lucknow	Dermatology	20	39	2023-05-05	2023-05-10	37856.70
29	Healing Touch	Pune	Cardiology	46	55	2023-09-25	2023-09-29	25298.50
30	City Hospital	Hyderabad	Dermatology	7	88	2023-05-12	2023-05-21	28558.23
31	Global Medicare	Chennai	Dermatology	36	69	2023-09-19	2023-09-20	44452.26
32	Healing Touch	Mumbai	Neurology	20	94	2023-08-21	2023-08-28	49955.41
33	Green Valley Hospital	Jaipur	Dermatology	38	139	2023-07-01	2023-07-15	14118.62
34	Global Medicare	Jaipur	Urology	38	62	2023-06-07	2023-06-17	22074.89
35	Global Medicare	Kolkata	Orthopedics	22	133	2023-02-13	2023-02-28	18795.69
36	Wellness Clinic	Jaipur	Neurology	47	84	2023-01-20	2023-01-26	26646.52
37	Fortis Care	Jaipur	Oncology	15	40	2023-10-23	2023-10-26	37057.29
38	Healing Touch	New Delhi	General Medicine	46	23	2023-05-27	2023-05-29	40771.84
39	City Hospital	Kolkata	Urology	6	166	2023-12-16	2023-12-25	40731.29
40	Wellness Clinic	Lucknow	Urology	37	85	2023-06-09	2023-06-16	4872.04
41	Metro Hospital	Lucknow	Neurology	22	167	2023-12-29	2024-01-10	44531.22
42	Apollo Health	Chennai	ENT	8	61	2023-09-10	2023-09-15	42769.14
43	Fortis Care	Bengaluru	Gynecology	30	69	2023-11-13	2023-11-26	14701.02
44	City Hospital	Jaipur	Gynecology	11	22	2023-05-15	2023-05-18	48466.08
45	Sunrise Medical	Bengaluru	ENT	37	26	2023-07-10	2023-07-24	25367.78
46	Green Valley Hospital	Ahmedabad	Dermatology	15	112	2023-03-26	2023-03-28	32628.67
47	Heritage Hospital	Lucknow	Urology	30	161	2023-11-10	2023-11-25	10166.89
48	Green Valley Hospital	Bengaluru	Pediatrics	8	177	2023-08-13	2023-08-17	15913.48
49	Healing Touch	Hyderabad	Pediatrics	30	158	2023-11-13	2023-11-22	16757.51
50	Wellness Clinic	Ahmedabad	Neurology	27	65	2023-03-29	2023-04-08	35720.61
51	Green Valley Hospital	Jaipur	General Medicine	45	88	2023-06-21	2023-07-05	9438.26
52	Wellness Clinic	Lucknow	Cardiology	32	51	2023-12-30	2024-01-12	27223.46
53	Metro Hospital	Kolkata	Gynecology	10	94	2023-01-18	2023-02-02	35570.84
54	Healing Touch	Ahmedabad	Urology	13	75	2023-02-07	2023-02-13	26027.37
55	Sunrise Medical	Mumbai	Urology	23	109	2023-06-23	2023-07-01	36909.89
56	Fortis Care	Pune	Oncology	41	47	2023-03-25	2023-03-29	8519.11
57	Metro Hospital	Chennai	Pediatrics	9	134	2023-05-17	2023-05-31	48873.72
58	Metro Hospital	Hyderabad	General Medicine	49	42	2023-12-28	2024-01-07	31342.58
59	Global Medicare	Ahmedabad	Orthopedics	48	138	2023-01-18	2023-01-26	46741.91
60	Global Medicare	Bengaluru	ENT	42	58	2023-09-18	2023-09-19	26614.52
61	Healing Touch	Ahmedabad	Urology	21	50	2023-06-22	2023-06-27	10978.18
62	Global Medicare	New Delhi	ENT	21	66	2023-12-29	2024-01-13	31175.27
63	Sunrise Medical	Pune	Orthopedics	31	122	2023-11-10	2023-11-13	44290.86
64	Heritage Hospital	Bengaluru	ENT	26	85	2023-09-25	2023-09-29	7502.64

65	Metro Hospital	Lucknow	ENT	31	75	2023-06-06	2023-06-11	7121.37
66	Heritage Hospital	Jaipur	Urology	39	70	2023-07-28	2023-08-04	48241.00
67	City Hospital	Ahmedabad	Neurology	14	91	2023-07-20	2023-08-03	33446.24
68	Fortis Care	Bengaluru	ENT	14	62	2023-12-17	2023-12-30	13590.65
69	City Hospital	Chennai	Gynecology	28	84	2023-02-08	2023-02-22	20849.31
70	Green Valley Hospital	Hyderabad	Neurology	35	57	2023-08-09	2023-08-19	4388.33
71	Healing Touch	Pune	Pediatrics	5	41	2023-03-31	2023-04-05	23916.63
72	Heritage Hospital	Bengaluru	Urology	29	54	2023-10-22	2023-11-05	5084.11
73	Apollo Health	Ahmedabad	Neurology	28	102	2023-12-13	2023-12-20	23328.90
74	City Hospital	New Delhi	Pediatrics	41	27	2023-02-20	2023-03-06	33417.28
75	Healing Touch	Jaipur	Cardiology	6	110	2023-09-21	2023-10-06	36545.43
76	Sunrise Medical	Jaipur	Oncology	19	31	2023-07-01	2023-07-15	43687.01
77	Sunrise Medical	Kolkata	Cardiology	7	50	2023-08-02	2023-08-07	13902.24
78	Apollo Health	Mumbai	Neurology	32	56	2023-01-11	2023-01-26	22456.61
79	Metro Hospital	Chennai	Urology	28	154	2023-09-04	2023-09-19	23012.64
80	Heritage Hospital	Lucknow	Neurology	44	194	2023-08-18	2023-08-27	33963.65
81	Apollo Health	Hyderabad	Gynecology	40	150	2023-05-01	2023-05-02	9775.99
82	Metro Hospital	Ahmedabad	Urology	38	108	2023-04-08	2023-04-20	17211.57
83	Global Medicare	Ahmedabad	Urology	17	29	2023-01-16	2023-01-25	17999.48
84	Metro Hospital	Hyderabad	Dermatology	43	96	2023-02-04	2023-02-05	42883.43
85	Global Medicare	New Delhi	Cardiology	21	151	2023-06-07	2023-06-09	2854.22
86	Sunrise Medical	Jaipur	Neurology	40	88	2023-03-22	2023-03-25	8333.48
87	City Hospital	Ahmedabad	Urology	25	131	2023-06-30	2023-07-02	47678.73
88	Fortis Care	Ahmedabad	Dermatology	9	106	2023-11-14	2023-11-21	44526.27
89	Healing Touch	Pune	General Medicine	48	25	2023-10-16	2023-10-28	4514.63
90	Metro Hospital	Ahmedabad	Gynecology	23	20	2023-04-17	2023-04-22	25418.26
91	Global Medicare	Pune	Orthopedics	6	48	2023-07-11	2023-07-24	46666.48
92	Fortis Care	Hyderabad	ENT	19	152	2023-09-18	2023-09-19	39298.30
93	Sunrise Medical	New Delhi	General Medicine	24	43	2023-09-29	2023-10-01	45451.78
94	Global Medicare	Jaipur	Pediatrics	43	133	2023-01-27	2023-02-11	24556.18
95	Green Valley Hospital	Kolkata	Oncology	20	77	2023-03-07	2023-03-18	7146.07
96	Wellness Clinic	Bengaluru	Gynecology	28	126	2023-11-29	2023-12-10	21065.59
97	Heritage Hospital	Jaipur	General Medicine	26	169	2023-02-18	2023-02-19	41677.34
98	Healing Touch	Bengaluru	Oncology	40	193	2023-11-01	2023-11-15	47625.70
99	Metro Hospital	Lucknow	Neurology	42	163	2023-05-19	2023-05-26	8684.16
100	Healing Touch	Bengaluru	Urology	14	37	2023-12-09	2023-12-15	17213.05

1. **Total Number of Patients**
   Write an SQL query to find the total number of patients across all hospitals.

   ANSWER:
   select sum(patients_count) AS Total_patients from hospital ;

	total_patients bigint
1	9347

2. **Average Number of Doctors per Hospital**

   Retrieve the average count of doctors available in each hospital.

   ANSWER:

   select hospital_name,avg(doctors_count) AS Average_Doctor from hospital group by hospital_name;

Doctor/postgres@PostgreSQL 18

No limit

Data Output Messages Notifications

SQL

	hospital_name character varying (100)	average_doctor numeric
1	City Hospital	24.7000000000000000
2	Healing Touch	25.6428571428571429
3	Global Medicare	28.7272727272727273
4	Fortis Care	24.5454545454545455
5	Sunrise Medical	28.6250000000000000
6	Green Valley Hospital	28.8750000000000000
7	Heritage Hospital	24.6666666666666667
8	Metro Hospital	27.4545454545454545
9	Apollo Health	23.3636363636363636
10	Wellness Clinic	34.0000000000000000

3. Top 3 Departments with the Highest Number of Patients

Find the top 3 hospital departments that have the highest number of patients.

ANSWER:

select hospital_name ,patients_count from hospital order by patients_count DESC LIMIT 3;

Data Output Messages Notifications

SQL

	hospital_name character varying (100)	patients_count integer
1	Heritage Hospital	198
2	Heritage Hospital	194
3	Healing Touch	193

4. Hospital with the Maximum Medical Expenses

Identify the hospital that recorded the highest medical expenses.

ANSWER:

select hospital_name,medical_expenses from hospital order by medical_expenses DESC LIMIT 1;

Data Output Messages Notifications

SQL

	hospital_name character varying (100)	medical_expenses numeric (10,2)
1	Healing Touch	49955.41

5. **Daily Average Medical Expenses**

Calculate the average medical expenses per day for each hospital.

ANSWER:

select hospital_name,avg(medical_expenses) from hospital group by hospital_name;

Dashboard × Properties × SQL × Statistics × Dependencies × Dependents

Doctor/postgres@PostgreSQL 18

No limit

Data Output Messages Notifications

SQL

	hospital_name character varying (100)	avg numeric
1	City Hospital	34798.844000000000
2	Healing Touch	25155.575714285714
3	Global Medicare	29976.313636363636
4	Fortis Care	30697.941818181818
5	Sunrise Medical	29131.366250000000
6	Green Valley Hospital	20273.603750000000
7	Heritage Hospital	20766.558888888889
8	Metro Hospital	28616.913636363636
9	Apollo Health	25212.280909090909
10	Wellness Clinic	25066.831428571429

6. **Longest Hospital Stay**

Find the patient with the longest stay by calculating the difference between Discharge Date and Admission Date.

ANSWER:

SELECT admission_Date, discharge_Date, (discharge_date - admission_date) AS StayDuration FROM hospital ORDER BY StayDuration DESC LIMIT 1;

Data Output Messages Notifications

SQL

	admission_date date	discharge_date date	stayduration integer
1	2023-04-06	2023-04-21	15

7. **Total Patients Treated Per City**

Count the total number of patients treated in each city.

ANSWER:

select location,sum(patients_count) as Total_Patients from hospital group by location;

86	Sunrise Medical	Jaipur	Neurology	40	88	2023-03-22	2023-03-25	8333.48
87	City Hospital	Ahmedabad	Urology	25	131	2023-06-30	2023-07-02	47678.73
88	Fortis Care	Ahmedabad	Dermatology	9	106	2023-11-14	2023-11-21	44526.27
89	Healing Touch	Pune	General Medicine	48	25	2023-10-16	2023-10-28	4514.63
90	Metro Hospital	Ahmedabad	Gynecology	23	20	2023-04-17	2023-04-22	25418.26
91	Global Medicare	Pune	Orthopedics	6	48	2023-07-11	2023-07-24	46666.48
92	Fortis Care	Hyderabad	ENT	19	152	2023-09-18	2023-09-19	39298.30
93	Sunrise Medical	New Delhi	General Medicine	24	43	2023-09-29	2023-10-01	45451.78
94	Global Medicare	Jaipur	Pediatrics	43	133	2023-01-27	2023-02-11	24556.18
95	Green Valley Hospital	Kolkata	Oncology	20	77	2023-03-07	2023-03-18	7146.07
96	Wellness Clinic	Bengaluru	Gynecology	28	126	2023-11-29	2023-12-10	21065.59
97	Heritage Hospital	Jaipur	General Medicine	26	169	2023-02-18	2023-02-19	41677.34
98	Healing Touch	Bengaluru	Oncology	40	193	2023-11-01	2023-11-15	47625.70
99	Metro Hospital	Lucknow	Neurology	42	163	2023-05-19	2023-05-26	8684.16
							23-12-15	17213.05

Data Output Messages Notifications

SQL

	location character varying (100)	total_patients bigint
1	Jaipur	1505
2	Bengaluru	955
3	Lucknow	1264
4	Mumbai	483
5	New Delhi	421
6	Kolkata	596
7	Chennai	840
8	Pune	555
9	Hyderabad	1261
10	Ahmedabad	1467

8. Average Length of Stay Per Department

Calculate the average number of days patients spend in each department.

ANSWER:

SELECT department, AVG(discharge_date - admission_date) AS TotalStayDays FROM hospital GROUP BY department;

Data Output Messages Notifications

SQL

	department character varying (100)	totalstaydays numeric
1	Dermatology	5.6000000000000000
2	Oncology	8.1111111111111111
3	ENT	8.0833333333333333
4	Neurology	9.2500000000000000
5	Gynecology	7.6666666666666667
6	Pediatrics	9.1111111111111111
7	Orthopedics	7.1428571428571429
8	Cardiology	6.8571428571428571
9	Urology	8.7222222222222222
10	General Medicine	7.4285714285714286

9. Identify the Department with the Lowest Number of Patients

Find the department with the least number of patients.

ANSWER:

select department,min(patients_count) as MinPatient from hospital group by department ORDER BY MinPatient ASC LIMIT 1 ;

Data Output Messages Notifications

	department character varying (100)	minpatient integer
1	Gynecology	20

10.Monthly Medical Expenses Report

Group the data by month and calculate the total medical expenses for each month.

ANSWER:

select extract(MONTH FROM admission_date) As MONTH ,sum(medical_expenses) from hospital group by extract(MONTH FROM admission_date) Order by MONTH;

Data Output Messages Notifications

	month numeric	sum numeric
1	1	173971.54
2	2	301722.72
3	3	199247.42
4	4	88995.93
5	5	222986.72
6	6	165926.36
7	7	211527.13
8	8	181039.55
9	9	341284.23
10	10	158450.68
11	11	334370.33
12	12	337788.45